Networks, Phillips Curves, and Exchange Rate Policy: A Small Open Economy Extension of Rubbo (2024)

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Abstract

I extend Rubbo's (2024) closed-economy theory of Phillips curves in production networks to a small open economy (SOE) with an exchange-rate channel, and ask whether her "divine coincidence" result — the existence of a single inflation index whose targeting simultaneously closes the output gap — survives when the exchange rate becomes a second, independent cost-push channel alongside sectoral productivity. I show theoretically that it generically does not: the divine-coincidence index is the unique left null vector of a cost-push matrix that must additionally annihilate the exchange-rate pass-through vector, two conditions that a single vector can jointly satisfy only in the knife-edge case where pass-through is proportional to labor intensity. I take the model to three real national input–output calibrations (Chile, South Korea, Czechia) and compare float, peg, and managed-float exchange-rate regimes. Managed float dominates in every calibration; peg is the worst regime, and its loss is driven overwhelmingly (58–83% depending on the calibration of the debt-elastic risk-premium coefficient) by a purely financial risk-premium/uncovered-interest-parity shock rather than by the terms-of-trade channel emphasized in the classical Mundell–Fleming and Galí–Monacelli literature. An extensive robustness campaign — roughly 350 additional model solves spanning network density, network topology, uniform and sector-specific price rigidity, the interaction between the risk-premium shock and its closing-device elasticity, and alternative policy-rule designs — confirms the ranking survives essentially everywhere, while also uncovering that (i) a

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sufficiently aggressive plain inflation-targeting rule can rival the managed float's performance without any explicit exchange-rate term, and (ii) once policy is aggressive, targeting domestic (PPI) inflation outperforms targeting the theoretically "correct" divine-coincidence index. I discuss the paper's relation to the contemporaneous and independent Qiu, Wang, Xu and Zanetti (2026) result, and lay out the concrete steps — a fully disaggregated multi-sector calibration and an estimated (rather than assumed) risk-premium process chief among them — required before the quantitative ranking can be treated as more than suggestive.

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1 Introduction

Modern economies are not single-good economies. Firms buy inputs from other firms, and the resulting input–output (IO) network shapes how shocks propagate and how monetary policy should be conducted. Rubbo (2024) formalizes this insight for a closed economy: in an N-sector New Keynesian model with an arbitrary IO matrix and Calvo price-setting frequencies that differ across sectors, there exists a unique inflation index — weighted by each sector's Domar (total sales) weight and by its price stickiness — whose targeting delivers the "divine coincidence": zero cost-push inflation and a simultaneously closed output gap. This is a striking generalization of the one-sector New Keynesian benchmark, in which any targeting rule that stabilizes marginal cost also stabilizes the output gap. Rubbo's result says that this coincidence survives the introduction of arbitrarily rich production networks and heterogeneous price stickiness, provided the central bank targets the *right* index rather than a naive aggregate like CPI inflation.

This paper asks whether the divine coincidence survives a second, empirically central feature of essentially every economy outside the largest few: openness. A small open economy (SOE) imports intermediate inputs, its firms' marginal costs depend on the nominal exchange rate, and its central bank must additionally decide how to respond to exchange-rate movements — float, peg, or some managed combination of the two. I show that adding this single, minimal ingredient — an import cost share in every sector's marginal cost, priced at the law-of-one-price exchange rate — breaks the divine coincidence generically. The mechanism is transparent: the exchange rate is a *second* cost-push channel, exactly analogous to the sectoral productivity shocks that generate the first one. Rubbo's unique index is pinned down by requiring it to annihilate the productivity cost-push matrix; requiring it to *also* annihilate the exchange-rate pass-through vector is a second, generically independent linear restriction on the same weighting vector, which a vector of finite dimension can satisfy only in a non-generic, measure-zero case. Divine coincidence in the open economy is a knife-edge property, not a generic one.

This theoretical result immediately raises a quantitative question with obvious policy relevance: if no simple inflation index can costlessly close the output gap under a float, how much welfare is actually at stake across the natural menu of exchange-rate regimes — float, hard peg, and managed float — and through which shocks? I calibrate the model to real, disaggregated national input–output data for three small open economies (Chile, South Korea, and Czechia) and solve it in Dynare under each regime. The headline finding is that a managed float — a conventional Taylor rule augmented with a modest, positive response to the exchange rate — dominates both corners in every calibration, and that a hard peg is by a wide margin the worst regime. This ranking replicates the classical Mundell–Fleming/Galí-Monacelli conclusion that flexible exchange rates are stabilizing, but the *mechanism* is different and, I would argue, more interesting: decomposing each regime's welfare loss by shock shows that the peg's dominance is driven overwhelmingly by a purely financial shock — a debt-elastic risk premium on the economy's net foreign asset position, in the tradition of Schmitt-Grohé and Uribe's (2003) closing device — and not by the terms-of-trade channel that is the star of the classical open-economy literature. This is an uncomfortable result in one specific sense that I confront directly rather than hide: since the risk-premium shock's standard deviation is a modeling choice (set equal to every other shock's, for want of an external target), the entire "peg is dominated" headline finding is only as solid as that one calibration choice. I therefore subject it, and the closely related debt-elasticity coefficient governing how strongly the risk premium reacts to the net foreign asset position, to an extensive battery of robustness checks — described below — rather than treating the headline number as settled.

The theoretical contribution of this paper sits between two literatures that, until very recently, had not been brought together. The first is the closed-economy production-network literature (Rubbo, 2024; La'O and Tahbaz-Salehi, 2022; Pasten, Schoenle and Weber, 2020; Baqaee and Farhi, 2019, 2020), which shows that IO linkages and heterogeneous price stickiness matter enormously for how monetary policy should be conducted, but is silent on the exchange rate because it studies closed economies. The second is the open-economy New Keynesian literature descending from Galí and Monacelli (2005), Obstfeld and Rogoff (2000), and Corsetti, Dedola and Leduc (2010), which analyzes exchange-rate regimes and optimal monetary policy in richly specified open economies, but — with rare exceptions — collapses the entire domestic supply side to a single sector, so that it cannot speak to which *sectors* bear the cost of a given regime or how network structure shapes the regime ranking. A single contemporaneous paper, Qiu, Wang, Xu and Zanetti (2026), independently derives an open-economy generalization of Rubbo's index and reaches the same qualitative divine-coincidence-breakdown conclusion through a structurally different route (a static Galí–Monacelli trade block with expenditure-switching and profit channels, rather than this paper's dynamic import-cost-share pass-through vector). Their model, notably, has no financial account, no uncovered interest parity, and no exchange-rate-*regime* choice — their own stated top-priority extension is exactly the machinery (debt-elastic risk premium, near-unit-root net foreign assets) that this paper builds around, and that turns out to drive the entire quantitative result. Section 2 discusses this relationship, and the surrounding literature, in detail.

Given how much of the headline result rests on two specific, judgment-call calibration parameters (the risk-premium shock's standard deviation and its debt-elasticity coefficient), a substantial part of this paper — comprising roughly 350 additional Dynare solves beyond the three headline calibrations — is devoted to systematically stress-testing it. I vary: the density and the *shape* of the domestic production network (a stylized triangular network scaled continuously from zero to double density, plus two alternative topologies — a hub-and-spoke network with Manufacturing as the central node, and a one-directional supply chain); uniform and Services-specific price rigidity; the debt-elasticity coefficient of the risk-premium process, on its own, jointly with network density, and jointly with the risk-premium shock's own volatility; and the exact design of the monetary policy rule, including strict domestic- and consumer-price-inflation targeting and a coarse grid search over the Taylor rule's own coefficients. The central qualitative conclusion — Managed float dominates Float, which dominates Peg, in every single one of these variations — is remarkably stable. Two secondary findings from this exercise are, I think, worth flagging on their own terms: first, that a sufficiently aggressive plain inflation-targeting rule, with no explicit exchange-rate response at all, can rival the managed float's performance; and second, that once policy is that aggressive, targeting domestic (producer) price inflation outperforms targeting the theoretically privileged divine-coincidence index, reversing the ranking found under more moderate policy coefficients. Both nuance, without overturning, the paper's central welfare ranking, and both are flagged explicitly as preliminary rather than settled, since neither has yet been checked against a fully re-optimized managed-float rule at the same level of policy aggressiveness.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature in three strands: closed-economy production networks, open-economy monetary policy, and the empirical exchange-rate pass-through literature that disciplines the model's key structural parameters. Section 3 lays out the small open economy model. Section 4 derives the modified sectoral Phillips curve, states and proves the divine-coincidence breakdown result, and derives the welfare function used throughout the quantitative analysis. Section 5 describes the data and calibration strategy for the three country calibrations. Section 6 presents the headline quantitative results: the welfare ranking, its shock decomposition, a mechanism analysis of why the stickiest, least import-exposed sector ("Services") nonetheless bears the largest share of the welfare cost, and cross-country robustness. Section 7 presents the full robustness campaign. Section 8 discusses policy implications and the paper's remaining limitations. Section 9 concludes. Appendix A collects the detailed, step-by-step proofs of all lemmas and propositions stated in the main text.

2 Literature Review

This paper sits at the intersection of three literatures: production networks and monetary policy in closed economies, exchange-rate policy and the divine coincidence in (single-sector) open economies, and the empirical literature on how production networks shape international pass-through. I discuss each in turn, before turning to the single closest paper in the literature.

2.1 Production Networks and Monetary Policy in Closed Economies

Rubbo (2024) is the direct theoretical parent of this paper. In an N-sector economy with Calvo price-setting, heterogeneous sectoral stickiness $\hat{\delta}_i$, and an arbitrary IO matrix Ω, she shows three results this paper builds on directly. First, natural (flexible-price) output equals Domar-weighted total factor productivity (a version of Hulten's, 1978, theorem applied to the social planner's problem). Second, sectoral inflation rates satisfy a vector Phillips curve $\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t$, in which the cost-push matrix $\mathcal{V}$ has rows that sum to zero — only *relative*, not uniform, productivity shocks generate cost-push inflation — and is built from the rigidity-adjusted Leontief inverse $(I - \Omega\Delta)^{-1}$, so that stickier upstream sectors dampen the propagation of shocks by absorbing them into markups rather than passing them through as price changes. Third, and most importantly for this paper, there is a unique inflation index $\pi_t^{DC} = \sum_i w_i^{DC}\pi_{it}$, with weights proportional to $\lambda_i(1 - \hat{\delta}_i)/\hat{\delta}_i$ (Domar weight times a stickiness multiplier), whose own law of motion has no cost-push term at all: targeting it in a standard Taylor-type rule simultaneously delivers a zero output gap. Rubbo further shows that a naive CPI-targeting rule, which ignores this network structure, can generate welfare losses of 2.9–3.8% of GDP relative to the divine-coincidence optimum even in the closed economy — motivation for taking network structure seriously as a first-order input into monetary policy design, not a second-order refinement.

La'O and Tahbaz-Salehi (2022) reach a closely related conclusion from a different direction: using a non-parametric, sufficient-statistics approach (rather than Rubbo's log-linear approximation), they show that optimal monetary policy in a production-network economy depends only on sectoral sales shares, demand elasticities, and the degree of strategic complementarity in pricing, and that the welfare-relevant inflation target weights sectors by Domar weight and inverse markup sensitivity — not by consumption share, as a naive CPI target would. Pasten, Schoenle and Weber (2020) provide the quantitative counterpart, calibrating a multi-sector Calvo model to U.S. input–output and Calvo frequency data (from Nakamura and Steinsson, 2008) and showing that network linkages amplify the real effects of a monetary policy shock by a factor of roughly two to three relative to an equivalent one-sector economy, with sticky, high-Domar-weight (frequently downstream, service-sector-like) industries accounting for a disproportionate share of the amplification — a pattern this paper's own "Services" mechanism, discussed in Section 6, closely parallels in the open economy.

Underlying all of the above is the foundational network-macro literature: Hulten (1978) and Acemoglu, Carvalho, Ozdaglar and Tahbaz-Salehi (2012) establish that idiosyncratic sectoral shocks need not average out at the standard $1/\sqrt{N}$ rate once a fat-tailed Domar-weight distribution is present, so that individually small shocks to central ("hub") sectors can generate aggregate fluctuations. Baqaee and Farhi (2019, 2020) generalize Hulten's first-order result to second order, showing that the exact aggregate effect of sectoral distortions requires knowing not just Domar weights but the full pattern of substitution elasticities across sectors; the cross-sector welfare terms Φ_C, Φ_s that appear in Rubbo's Proposition 3, and in this paper's welfare function (Section 4), are precisely these Baqaee–Farhi operators. Bigio and La'O (2020) apply the same network-aggregation logic to arbitrary wedges (taxes, markups, financial frictions) rather than pure productivity shocks — the natural lens through which to view this paper's exchange-rate cost-push term, which is, formally, a heterogeneous wedge on each sector's imported-input cost.

2.2 Exchange-Rate Policy and the Divine Coincidence in Open Economies

The benchmark open-economy New Keynesian model is Galí and Monacelli (2005): a small open economy facing a large foreign economy, with households consuming a CES aggregate of home and foreign goods, firms using only labor (no intermediate inputs, let alone a production network), complete international financial markets, and the law of one price. In this environment, targeting domestic (PPI) inflation delivers the one-sector divine coincidence exactly as in a closed economy, with the flexible exchange rate acting as an automatic stabilizer of the terms of trade; targeting CPI inflation instead is strictly suboptimal because it forces the central bank to partially resist exchange-rate movements that would otherwise be efficient; and a hard peg is welfare-inferior because it surrenders monetary independence (the Mundell trilemma) in the face of domestic productivity shocks. Corsetti, Dedola and Leduc (2010) survey the conditions under which this benchmark divine coincidence survives richer environments — incomplete markets, pricing-to-market, local-currency pricing — and under which it breaks down; the production-network channel this paper introduces is absent from every model they survey.

A second relevant strand concerns the financial side of the small open economy. Schmitt-Grohé and Uribe's (2003) debt-elastic interest-rate premium is the standard closing device for models with incomplete asset markets and a stationary net-foreign-asset position; this paper adopts it directly, and much of the robustness analysis in Section 7 is devoted to stress-testing the model's sensitivity to its calibration. Fanelli and Straub (2021) provide the theoretical underpinning for treating foreign-exchange intervention as a genuine second policy instrument alongside the interest rate, showing formally that a central bank facing a UIP risk premium should lean against the component of exchange-rate movements that generates inefficient real effects while allowing the fundamentals-driven component to pass through — exactly the logic behind this paper's "managed float" regime, which augments a standard Taylor rule with a response to the exchange rate. Gopinath and Itskhoki's (2022) dominant-currency-paradigm literature is a natural extension this paper does not pursue: the baseline model assumes producer-currency pricing throughout, and allowing a fraction of imports to be invoiced in a third (dominant) currency would alter the exchange-rate pass-through vector in ways left for future work. Schmitt-Grohé and Uribe (2016) show that adding downward nominal wage rigidity to a pegged small open economy generates large, persistent involuntary unemployment following external shocks — since this paper's model has fully flexible wages, this suggests the peg's welfare cost documented here is, if anything, a conservative lower bound.

2.3 Production Networks and International Pass-Through

A largely empirical literature establishes that production-network linkages are quantitatively important for how prices and shocks transmit across borders, disciplining this paper's calibration of the exchange-rate pass-through vector Γ. Auer, Levchenko and Sauré (2019) show, using international input–output data for forty countries, that a one percent increase in a trading partner's producer price inflation, weighted by bilateral IO exposure, raises domestic producer price inflation by 0.15–0.25 percent above and beyond standard trade-weighted exchange-rate effects, with upstream (supplier-side) linkages mattering more than downstream ones — direct motivation for this paper's $\Gamma \Delta e_t$ cost-push term, and for using disaggregated IO data rather than aggregate trade shares to calibrate it. Amiti, Itskhoki and Konings (2014) show at the firm level that import intensity is the key determinant of incomplete exchange-rate pass-through to export prices (a natural hedge: depreciation raises both a firm's foreign revenue and its imported-input costs), consistent with this paper's assumption that Γ_i is proportional to sector i's import cost share, amplified through the network. Bems and Johnson (2017) develop "value-added exchange rates," showing that standard trade-weighted real effective exchange rates are systematically biased for economies deeply embedded in global value chains — the empirical counterpart of this paper's insight that a sector's *effective* exposure to the exchange rate depends on the entire chain of its suppliers' import intensities, not just its own. Nakamura and Steinsson (2008) and Carvalho (2006) document the extensive cross-sector heterogeneity in price-adjustment frequency that gives Rubbo's, and this paper's, $\hat{\delta}_i$-weighted objects their empirical bite; without this heterogeneity, the divine-coincidence index would collapse to a simple Domar-weighted (or, under Cobb–Douglas consumption, consumption-weighted) average and the distinction this paper studies would be moot.

2.4 The Closest Paper: Qiu, Wang, Xu and Zanetti (2026)

The single closest paper in the literature is Qiu, Wang, Xu and Zanetti (2026, *Journal of Monetary Economics*), an independent and contemporaneous extension of Rubbo's framework to the open economy. Their model combines Galí–Monacelli's one-sector open-economy trade block with Rubbo's multi-sector production network, calibrated to the World Input-Output Database (43 countries by 56 sectors). They derive an open-economy generalization of the divine-coincidence index in which the output-gap weight on each sector decomposes into a CPI channel, an expenditure-switching channel, and a profit channel (the latter two specific to the open economy), and show — as this paper does, through a structurally different mechanism — that full divine coincidence (zero welfare loss, not merely a zero output gap) fails even after their generalized index is targeted, because the within- sector, across-sector, and cross-border misallocation terms are not simultaneously zeroed by output-gap targeting alone. Calibrated to WIOD, they show output-gap targeting is close to the constrained-optimal policy and dominates naive PPI or CPI targeting by a margin that rises with an economy's openness.

Three features distinguish this paper from theirs, and explain why their qualitative conclusion and this paper's are corroborating rather than redundant. First, their model is static (one period, no dynamics or persistence), while this paper's central mechanism — the interaction between a persistent, near-unit-root net-foreign-asset process and a debt-elastic risk premium — is inherently dynamic and could not be represented in their framework. Second, their model assumes balanced trade with no uncovered interest parity, no net foreign assets, and no financial account at all; the exchange rate is pinned down period-by-period by a static trade-balance condition. Third, and consequently, their model has no exchange-rate-*regime* choice: money supply targets an inflation index and the exchange rate floats implicitly throughout, so a peg-versus-float-versus- managed-float welfare comparison, this paper's central quantitative exercise, is outside their scope. Tellingly, the authors themselves name "relax[ing] the assumption of financial autarky" as their first, top-priority item for future research — precisely the piece of machinery this paper is built around. This paper's contribution, then, is best read as picking up exactly where Qiu, Wang, Xu and Zanetti's own stated research agenda leaves off, and the finding that the risk-premium/UIP channel — not the terms-of-trade or expenditure-switching channels their model emphasizes — turns out to dominate the exchange-rate-regime welfare ranking once that machinery is added.

A related, purely empirical paper is Silva (2024), who applies a production-network SOE framework to Chilean and UK data to explain the 2021–2022 inflation surge via a CPI-elasticity decomposition distinguishing direct from network-inherited import and export exposure. Silva's paper is positive rather than normative — it does not compare monetary or exchange-rate regimes — but its Chilean application provides a useful external check on the plausibility of this paper's own Chile-calibrated pass-through magnitudes, and the general finding that indirect (network-inherited) exposure matters quantitatively for inflation dynamics is the same mechanism this paper's "Why Services?" analysis (Section 6) documents in the welfare, rather than the positive-inflation, domain.

3 Model

The model extends Rubbo's (2024) closed-economy N-sector New Keynesian production-network economy to a small open economy with an imported-input sector, incomplete international financial markets, and a menu of exchange-rate policy regimes. I present the environment in full; the log-linearization and all formal results are proved in Appendix A.

3.1 Households

A representative household maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \rho^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right] \tag{1}$$

subject to the flow budget constraint

$$P_t^C C_t + B_{t+1} + \mathcal{E}_t B_{t+1}^* \leq W_t L_t + \Pi_t - T_t + (1+i_t) B_t + \mathcal{E}_t (1+i_t^*) \, \mathrm{RP}_t \, B_t^* \tag{2}$$

where C_t is a consumption aggregate of domestic and foreign varieties, B_t is a domestic bond (in zero net supply in equilibrium, but included explicitly rather than assumed away), B_t^* is the economy's net foreign asset position (in units of foreign currency), $\mathcal{E}_t$ is the nominal exchange rate (domestic currency per unit of foreign currency, so that an increase is a depreciation), and RP_t is a time-varying risk-premium wedge on foreign-currency returns, discussed below. Consumption C_t is a CES aggregate of a domestic consumption bundle C_t^H (itself Cobb–Douglas across the N domestic sectors with shares $\boldsymbol{\beta}^H$) and a foreign consumption composite C_t^F (a single Armington import aggregate, not sector-indexed),

$$C_t = \left[\omega^{1/\eta}(C_t^H)^{(\eta-1)/\eta} + (1-\omega)^{1/\eta}(C_t^F)^{(\eta-1)/\eta}\right]^{\eta/(\eta-1)} \tag{3}$$

with η the (CES) elasticity of substitution between home and foreign consumption bundles and ω the home bias parameter. The household's intratemporal and intertemporal optimality conditions are the standard labor-supply condition $W_t/P_t^C = C_t^\gamma L_t^\varphi$ and Euler equation, exactly as in the closed economy; the presence of the foreign bond additionally delivers an uncovered interest parity (UIP) condition, given below.

3.2 Firms, Technology, and the Production Network

There are N domestic industries (in the quantitative model, $N = 3$: Resource, Manufacturing, and Services), each populated by a unit mass of monopolistically competitive firms. A firm f in sector i produces using labor, domestic intermediate inputs from every sector j, and imported inputs,

$$Y_{ift} = A_{it}\, L_{ift}^{\alpha_i} \prod_{j=1}^{N} X_{ijft}^{\Omega_{ij}^H}\, M_{ift}^{\Omega_i^F}, \qquad \alpha_i + \sum_j \Omega_{ij}^H + \Omega_i^F = 1, \tag{4}$$

Cobb–Douglas across all of labor, the N domestic inputs, and the imported-input composite M_{ift} (priced, under the law of one price and producer-currency pricing, at $P_t^F = \mathcal{E}_t P_t^*$). Ω^H is the $N \times N$ domestic input–output matrix exactly as in Rubbo (2024); $\boldsymbol{\Omega}^F$ is the new object this paper introduces, the vector of sector-level imported-input cost shares. Within-sector demand and the Calvo price-setting mechanism are unchanged from the closed economy: a fraction δ_i of firms in sector i reset their price each period, generating, after log-linearization, an effective Calvo parameter $\hat{\delta}_i \equiv \delta_i(1-\rho(1-\delta_i))/(1-\rho\delta_i(1-\delta_i))$ exactly as in Rubbo (2024). Cost minimization gives sector i's log-linearized marginal cost as

$$mc_{it} = \alpha_i w_t + \sum_j \Omega_{ij}^H\, p_{jt} + \Omega_i^F\,(p_t^* + e_t) - \log A_{it} \tag{5}$$

the only modification to Rubbo's marginal cost equation being the new term $\Omega_i^F(p_t^* + e_t)$: the nominal exchange rate $e_t \equiv \log \mathcal{E}_t$ enters every sector's marginal cost in proportion to its imported-input cost share, exactly as sectoral productivity does, but with an opposite sign and through a different cost share vector ($\boldsymbol{\Omega}^F$ rather than $-\log \mathbf{A}_t$).

3.3 Foreign Sector, UIP, and the Risk Premium

The economy is small: foreign variables (P_t^*, foreign demand D_t^*, the foreign interest rate i_t^*) are exogenous. Uncovered interest parity, augmented with a debt-elastic risk premium in the spirit of Schmitt-Grohé and Uribe (2003), reads

$$(1 + i_t) = (1 + i_t^*) \Big(1 - \Psi\big(B_t^* - \bar{B}^*\big)\Big) \, \mathrm{RP}_t \, \frac{\mathbb{E}_t \mathcal{E}_{t+1}}{\mathcal{E}_t} \tag{6}$$

where $\Psi > 0$ is the debt-elasticity coefficient (governing how strongly the domestic interest rate must compensate foreign lenders as the economy's net foreign asset position falls below its steady- state target $\bar{B}^*$, the standard device restoring stationarity to an otherwise near-random-walk net foreign asset position) and RP_t is an exogenous, stochastic risk-premium shock, $\log \mathrm{RP}_t = \rho_{RP} \log \mathrm{RP}_{t-1} + \varepsilon_t^{RP}$. This risk-premium shock plays the central quantitative role in the paper's results and is subjected to extensive robustness analysis in Section 7. The economy's net foreign asset position evolves according to the balance of payments,

$$B_t^* = \frac{1 + i_{t-1}^*}{\Pi_t^*} \, \mathrm{RP}_{t-1} \, B_{t-1}^* + \frac{\mathrm{TB}_t}{\mathcal{E}_t P_t^*}, \tag{7}$$

where TB_t is the nominal trade balance (exports minus imports). Exports are driven by foreign demand and the relative price of domestic to foreign goods, $EX_t = \kappa_{EX} D_t^* (P_t^H / P_t^{FH})^{-\theta^*}$, with θ^* the (CES) elasticity of foreign demand for domestic goods.

3.4 Monetary Policy Regimes

I compare three monetary policy regimes, the paper's central object of comparison. Under **float**, the central bank follows a conventional Taylor rule responding to the divine-coincidence index and the output gap,

$$\log(I_t / I^*) = \phi_\pi \pi_t^{DC} + \phi_y \, \tilde{y}_t, \tag{Float}$$

with no direct response to the exchange rate. Under **peg**, the exchange rate is fixed, $S_t \equiv 1$, and the interest rate becomes the residual variable pinned down by the UIP equation (6) — the standard Mundell trilemma trade-off, surrendering monetary independence in exchange for exchange-rate stability. Under **managed float**, the Taylor rule is augmented with an explicit response to the exchange rate,

$$\log(I_t/I^*) = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log S_t, \tag{Managed}$$

with $\phi_s > 0$ a partial FX-stabilization weight. Section 7 additionally considers strict domestic (PPI) and consumer-price (CPI) inflation targeting, and a coarser grid search over (ϕ_π, ϕ_y) themselves, as further regime-rule variants.

3.5 Equilibrium

An equilibrium is a sequence of prices and quantities such that: households optimize subject to their budget constraint; firms in every sector minimize cost and, when given the opportunity, reset prices optimally under Calvo pricing; goods markets clear sector by sector, $Y_{it} = C_{it}^H + \sum_j X_{jit} + EX_{it}$, where EX_{it} is sector i's share of aggregate exports (allocated, in the baseline calibration, via household consumption shares $\boldsymbol{\beta}^H$, and via sector-specific export intensities in the robustness extension discussed in Section 7); the labor market clears, $L_t = \sum_i L_{it}$; the UIP condition (6) holds under float and managed float, or the peg $S_t \equiv 1$ replaces it; and the monetary authority sets the interest rate according to its regime-specific rule. GDP is a reporting identity, $GDP_t = P_t^C C_t + P_t^H EX_t - P_t^{FH} IM_t$, and is deliberately not a market-clearing condition.

4 Theoretical Results

This section states the paper's central theoretical results: the modified sectoral Phillips curve with an exchange-rate cost-push term, the generic breakdown of the divine coincidence, and the welfare function used throughout the quantitative analysis. Full, step-by-step proofs — following exactly the same log-linearization and Sherman–Morrison-identity techniques Rubbo (2024) uses in the closed economy — are given in Appendix A; here I state results and give the essential economic content of each proof.

4.1 The Closed-Economy Benchmark

Three results from Rubbo (2024), restated here because the open-economy results are direct generalizations, or direct violations, of them.

Lemma 1 (Natural output)**.** *Natural (flexible-price) output equals Domar-weighted total factor productivity,* $y_{t,\text{nat}} = \frac{1+\varphi}{\gamma+\varphi}\boldsymbol{\lambda}^\top \log \mathbf{A}_t$*, where* $\boldsymbol{\lambda}^\top = \boldsymbol{\beta}^\top (I - \Omega^H)^{-1}$ *are Domar weights.*

Proposition 1 (Sectoral Phillips curve, closed economy)**.** *Sectoral inflation satisfies*

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\,\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t, \tag{8}$$

where $\mathcal{B} \equiv \Delta(I - \Omega^H\Delta)^{-1}\boldsymbol{\alpha}/\kappa$*,* $\kappa \equiv 1 - \boldsymbol{\beta}^\top\Delta(I - \Omega^H\Delta)^{-1}\boldsymbol{\alpha}$*, and* $\mathcal{V}$ *is a cost-push matrix, built from the same rigidity-adjusted Leontief inverse* $(I - \Omega^H\Delta)^{-1}$*, whose rows sum to zero:* $\mathcal{V}\mathbf{1} = \mathbf{0}$*.*

Proposition 2 (Divine coincidence, closed economy)**.** *There is a unique (up to normalization) inflation index* $\pi_t^{DC} = \boldsymbol{\phi}^{DC\top}\boldsymbol{\pi}_t$*,* $\boldsymbol{\phi}_i^{DC} \propto \lambda_i(1 - \hat{\delta}_i)/\hat{\delta}_i$*, that lies in the left null space of* $\mathcal{V}$ *and therefore satisfies a cost-push-free Phillips curve*

$$\pi_t^{DC} = \rho\mathbb{E}\pi_{t+1}^{DC} + \kappa_{DC}\tilde{y}_t. \tag{9}$$

Targeting π_t^{DC} *simultaneously delivers a zero output gap.*

Uniqueness in Proposition 2 follows because $\mathcal{V}\mathbf{1} = \mathbf{0}$ implies $\mathcal{V}$ has rank at most $N-1$, so its left null space is generically exactly one-dimensional. This one fact is the fulcrum on which the entire open-economy result in Section 4.3 turns.

4.2 The Modified Phillips Curve with an Exchange-Rate Channel

Substituting the open-economy marginal cost (5) into the same derivation that delivers Proposition 1 (Appendix A, Steps 1–7, unchanged in every step except the marginal cost equation itself) gives the paper's central reduced-form object.

Proposition 3 (Sectoral Phillips curve, open economy)**.** *Sectoral inflation satisfies*

$$\boxed{\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\,\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\,\Delta e_t} \tag{10}$$

where $\mathcal{B}$ *and* $\mathcal{V}$ *are exactly as in the closed economy (Proposition 1), and the new* ***exchange-rate pass-through vector*** *is*

$$\Gamma \equiv \frac{\Delta(I - \Omega^H\Delta)^{-1}\boldsymbol{\Omega}^F}{\kappa} = \frac{\tilde{A}\,\boldsymbol{\Omega}^F}{\kappa}, \qquad \tilde{A} \equiv \Delta(I - \Omega^H\Delta)^{-1}. \tag{11}$$

Equation (10) is the SOE analogue of Rubbo's Proposition 2, and its economic content is the paper's central mechanism in one line: the exchange rate enters every sector's Phillips curve exactly like a (negative) productivity shock, weighted by the *same* rigidity-adjusted network amplifier $\tilde{A} = \Delta(I - \Omega^H \Delta)^{-1}$ that governs productivity pass-through, but applied to the imported-input cost-share vector $\boldsymbol{\Omega}^F$ rather than to $-\log \mathbf{A}_t$. A sector's exposure to the exchange rate is therefore not simply its own import share Ω_i^F: it also inherits exposure from every supplier, weighted by how quickly that supplier resets its own price. A sector that imports nothing directly but buys heavily from an import-intensive supplier can be substantially exposed to the exchange rate purely through the network — the mechanism examined in detail in Section 6's "Why Services?" analysis.

4.3 Breakdown of the Divine Coincidence

Proposition 4 (Generic breakdown of the divine coincidence)**.** *Under a float, an inflation index $\boldsymbol{\phi}^\top \boldsymbol{\pi}_t$ has no cost-push term if and only if $\boldsymbol{\phi}^\top \mathcal{V} = \mathbf{0}^\top$* and *$\boldsymbol{\phi}^\top \Gamma = 0$. The first condition alone pins $\boldsymbol{\phi}$ down uniquely (Proposition 2); the second is then an* additional*, generically independent scalar restriction on the same, already-determined vector. Except in the knife-edge case $\Gamma \propto \mathcal{B}$ (exchange-rate pass-through exactly proportional to labor intensity), no inflation index exists that simultaneously eliminates both cost-push channels. The divine coincidence is a knife-edge, not a generic, property of open production networks.*

The proof (Appendix A) is almost immediate given Proposition 2: the closed-economy divine-coincidence weights $\boldsymbol{\phi}^{DC}$ are the *unique* (one-dimensional) left null vector of $\mathcal{V}$, so there is no freedom left to additionally satisfy $\boldsymbol{\phi}^{DC\top}\Gamma = 0$ unless this happens to hold already — a single additional scalar equation in zero remaining free parameters, generically inconsistent. The intuition is that Rubbo's closed economy has exactly one cost-push channel (productivity), so pinning down the unique index that annihilates it exhausts every degree of freedom available; adding the exchange rate as a *second*, generically linearly independent cost-push channel leaves no further freedom to annihilate it as well. This is the SOE analogue of the intuition behind the Tinbergen counting argument in Fanelli and Straub (2021): one instrument (or, here, one inflation index) cannot generally achieve two independent targets.

Remark 1 (Independent corroboration)**.** *Qiu, Wang, Xu and Zanetti (2026) reach the* *same qualitative conclusion — divine coincidence fails in a multi-sector open economy — through a structurally different route: rather than a single import-cost-share pass-through vector Γ, they derive a three-channel open-economy output-gap weight (CPI, expenditure-switching, profit channels) from a Galí–Monacelli-style trade block with balanced trade and export taxes, and show that output-gap targeting closes the gap but does not zero the within-sector, across-sector, and cross-border misallocation terms simultaneously. That two independent derivations, built on genuinely different modeling primitives, arrive at the same qualitative conclusion is worth treating as corroboration of the underlying economics rather than as redundant results.*

4.4 Welfare

Proposition 5 (Welfare, open economy)**.** *To second order, the welfare loss relative to the efficient (flexible-price) allocation is*

$$\mathbb{W} = \frac{1}{2}\,\mathbb{E}\Big[\underbrace{(\gamma+\varphi)\tilde{y}_t^2}_{\textit{output gap}} + \underbrace{\sum_i \kappa_i\,\pi_{it}^2}_{\textit{within-sector}} + \underbrace{\Phi_C(\boldsymbol{\mu}_t,\boldsymbol{\mu}_t) + \sum_s \lambda_s \Phi_s(\boldsymbol{\mu}_t,\boldsymbol{\mu}_t)}_{\textit{cross-sector}}\Big], \qquad \kappa_i \equiv \lambda_{D,i}\,\varepsilon_i\,\frac{1-\hat{\delta}_i}{\hat{\delta}_i}, \tag{12}$$

where $\boldsymbol{\mu}_t$ is the vector of sectoral log markup gaps and Φ_C, Φ_s are Baqaee–Farhi (2019) cross-sector substitution operators evaluated at the model's consumption and production elasticities of substitution.

The output-gap and within-sector terms are unchanged in form from the closed economy; the exchange rate enters welfare only indirectly, through its effect on the equilibrium paths of $\tilde{y}_t$ and $\boldsymbol{\pi}_t$ under each policy regime, plus a direct cross-sector misallocation term $\Phi_C(\tilde{\boldsymbol{\chi}}_t, e_t\boldsymbol{\Omega}^F)$ arising because different sectors face different import cost shocks, distorting relative prices even absent any within-sector dispersion. Because this model's cross-sector aggregators (household consumption across domestic sectors, and each firm's input bundle across domestic and imported inputs) are Cobb–Douglas by construction, the Allen–Uzawa elasticity of substitution between any pair of their arguments is exactly one, and the cross-sector terms Φ_C, Φ_s collapse to a weighted variance of the markup-gap vector, with only the outer consumption nest (home versus foreign consumption, $\eta = 1.5$) contributing a genuine non-unitary elasticity. Numerically (Appendix A, Remark 2), this cross-sector term adds a modest 3.3–8.4% on top of the output-gap-plus-within-sector total, depending on the regime, without affecting the regime ranking; the headline welfare numbers reported in Section 6 use the output-gap-plus-within-sector total (equation (12)'s first two terms) as the primary object of comparison, consistent with how every robustness exercise in Section 7 is computed, and I flag explicitly wherever the cross-sector term is or is not included.

5 Data and Calibration

I calibrate the model to three small open economies with genuinely disaggregated national input–output data: Chile, South Korea, and Czechia. Chile is the primary quantitative calibration used throughout the paper's main results and robustness analysis; Korea and Czechia serve as out-of-sample checks on whether the headline welfare ranking is a Chile-specific artifact or a more general property of the model (Section 6, cross-country robustness).

5.1 Input–Output Data and Sector Aggregation

For each country, I take the national supply-and-use (or, for Chile, Cuadro de Origen y Recursos, CdeR) tables and collapse the native sectoral classification (12 sectors for Chile, using Banco Central de Chile's 2018 CdeR tables at 2023 reference prices) to three broad sectors — Resource, Manufacturing, and Services — chosen to span a wide range of import intensity and price stickiness while keeping the calibrated model computationally tractable for the extensive robustness campaign in Section 7. From each country's IO table I compute: the domestic IO matrix Ω^H (each sector's cost share purchased from each domestic sector); the imported-input cost share vector $\boldsymbol{\Omega}^F$; the labor cost share $\boldsymbol{\alpha}$; household consumption shares $\boldsymbol{\beta}^H$; and each sector's export intensity (exports as a share of own gross output), used in the sector-specific export-reallocation robustness check of Section 7.

Table 1 reports the resulting Chile calibration — the parameterization used for every result in Sections 6 and 7 unless stated otherwise — along with the derived network objects (Domar weights $\lambda_{D,i}$, total import centrality M_i, and the divine-coincidence weights w_i^{DC}) used throughout the paper.

Table 1: Chile calibration: primitives and derived network objects

	Resource (1)	Manufacturing (2)	Services (3)
Labor share α_i	0.5026	0.3589	0.6035
Import share Ω_i^F	0.0767	0.1945	0.0704
Consumption share β_i^H	0.0265	0.2294	0.7441
Export share of own output	0.6019	0.1797	0.0357
Calvo reset probability δ_i	0.90	0.31	0.16
Domar weight $\lambda_{D,i}$	0.0718	0.3820	1.1084
Total import centrality M_i	0.1547	0.2847	0.1189
Effective Calvo persistence $\hat{\delta}_i$	0.8902	0.1246	0.0311
DC index weight w_i^{DC}	0.0002	0.0720	0.9277

Domestic IO matrix (rows buy from columns, Resource/Manufacturing/Services order): $\Omega^H = \begin{pmatrix} 0.0750 & 0.1526 & 0.1932 \\ 0.0991 & 0.2022 & 0.1453 \\ 0.0018 & 0.0581 & 0.2661 \end{pmatrix}$.

5.2 Calvo Frequencies

Sector-level Calvo reset probabilities δ_i are not identifiable from input–output data alone and are taken from the price-stickiness literature (Dhyne et al., 2005, the ECB's Inflation Persistence Network; cross-checked against Nakamura and Steinsson, 2008), assigning Resource the highest reset probability (0.90, reflecting globally-priced commodity-like goods), Manufacturing an intermediate value (0.31), and Services the lowest (0.16, reflecting the well-documented empirical stickiness of service prices). The resulting effective Calvo persistence $\hat{\delta}_i$ (Table 1) is markedly lower for Services (0.031) than for Resource (0.890): a one percent shock to Services' marginal cost translates into roughly thirty times less inflation, and correspondingly more absorbed into markup dispersion, than the same shock in the Resource sector — the origin of Services' outsized welfare weight κ_i discussed in Section 6.

5.3 Preference and Technology Parameters

The remaining parameters — the coefficient of relative risk aversion $\gamma = 1$, the inverse Frisch elasticity of labor supply $\varphi = 2$, the within-sector elasticity of substitution $\varepsilon = 8$, the discount factor $\beta = 0.99$, the home-foreign consumption elasticity $\eta = 1.5$, the export-demand elasticity $\theta^* = 2.0$, and the debt-elasticity coefficient of the risk premium $\Psi = 0.020$ — are literature defaults rather than estimates specific to this setting, and are flagged as such in the discussion of the paper's remaining limitations (Section 8). All exogenous shocks (sectoral productivity, import price, foreign demand, export price/terms-of-trade, and the risk premium itself) are calibrated to a common one percent standard deviation, in the absence of an external target for the risk-premium shock specifically — a calibration choice whose consequences for the headline result are the subject of extensive robustness analysis in Section 7.

6 Quantitative Results

6.1 Headline Welfare Ranking

Table 2 reports the model's central quantitative result: welfare loss, in units of 10^{-4} (i.e., basis points of steady-state consumption-equivalent loss times 100), for each of the three monetary/exchange-rate regimes, computed from equation (12)'s output-gap-plus- within-sector terms using the order-1 (log-linear) solution's unconditional variances.

Table 2: Headline welfare loss by regime, real Chile calibration ($\times 10^{-4}$)

	Float	Managed Float	Peg
Welfare loss ($\times 10^{-4}$)	25.47	10.17	102.05
Relative to Managed	2.5×	1× (best)	10.0×

Managed float dominates both corner regimes by a wide margin; Peg is roughly four times worse than Float and ten times worse than Managed float. This is, superficially, the same ranking the classical Mundell–Fleming/Galí–Monacelli literature would predict — flexible exchange rates are stabilizing, hard pegs are costly — but Section 6's shock decomposition shows the mechanism generating this ranking is not the one that literature emphasizes.

Including the cross-sector welfare term of equation (12) (Section 4, Remark on Cobb–Douglas cross-sector aggregators) raises every regime's loss modestly — 3.3% for Float, 8.4% for Managed, and 4.5% for Peg — to totals of 26.31, 11.03, and 106.62 ($\times 10^{-4}$) respectively, without affecting the ranking; unless stated otherwise, all subsequent tables in this paper use the output-gap-plus-within-sector total for direct comparability with the extensive robustness campaign of Section 7, none of which recomputes the cross-sector term at every grid point.

6.2 Shock Decomposition: The Risk-Premium Channel

Table 3 decomposes each regime's total welfare loss by the shock responsible for it, using Dynare's own forecast-error variance decomposition. The result is stark: the risk-premium shock ε_t^{RP} — a purely financial, non-import shock entering only through the UIP condition (6) — accounts for 75% of Peg's total welfare loss at the model's baseline calibration, far exceeding the terms-of-trade (export-price) channel that is the star of the classical open-economy literature (a mere 2.16 out of 102.05, or roughly 2%, of Peg's total loss).

Table 3: Share of each regime's welfare loss attributable to the risk-premium shock (%)

	Float	Managed	Peg
Risk-premium shock's % of total loss	$\approx$11	$\approx$8	**75**

This finding is quantitatively central to the paper, and uncomfortable in one specific sense that I confront directly: Section 7's risk-premium volatility sweep shows that with the risk-premium shock switched off entirely, Peg is not dominated at all — it is essentially tied with Float, fractionally *better*. The entire "Peg is worst" headline result therefore depends on the risk-premium shock existing at roughly its calibrated size, a calibration choice (equal standard deviation to every other shock, for lack of an independently estimated target) rather than an externally disciplined one. I return to this in Section 7, where I show the qualitative ranking survives even substantial variation in both the shock's size and the closing device's debt-elasticity coefficient, while being explicit that the *margin* of Peg's dominance is highly sensitive to both.

6.3 Why Services? A Sectoral Mechanism Decomposition

A natural puzzle, raised directly in early feedback on this project, is why the Services sector — the least import-exposed sector in the calibration (Table 1, lowest direct Ω_i^F) and, on that basis alone, the sector one might expect to be most insulated from exchange-rate-driven welfare costs — in fact absorbs the largest share of welfare loss in every regime. Three candidate explanations were raised, and all three turn out to be real and compounding rather than competing.

Indirect exposure via the network. Total import centrality M_i (equation (11)'s numerator, up to normalization) splits into a direct part (Ω_i^F, imports the sector buys itself) and an indirect part inherited via domestic suppliers. For Services, 41% of its total import centrality is indirect — almost as large a share as Resource's 50%, despite Services' total exposure being the smallest of the three sectors in levels. Concretely, Services buys 5.8% of its cost from Manufacturing, the most import-intensive sector in the economy; whatever exchange-rate cost-push hits Manufacturing directly is inherited by Services through this single supply link, and because Services cannot reprice quickly (its effective Calvo persistence $\hat{\delta}_3 = 0.031$ is an order of magnitude lower than Resource's), whatever cost-push it inherits this way does not clear within a quarter — it lingers, making Services' inflation the single most persistent series in the model (autocorrelation around 0.82).

Which channel actually dominates Services' own variance? Dynare's own variance decomposition of Services' output gap shows the import-price/network channel just described is real but *secondary* — explaining only 2–22% of Services' output-gap variance depending on the regime — while the purely aggregate, non-import risk-premium shock dominates: 63% under Float, 78% under Peg.

Why does a non-import shock land so heavily on Services specifically? Because of its Domar weight, $\lambda_{D,3} = 1.108$ — more than double the next-largest sector's — which reconciles the apparent tension: 92% of this weight is already present at zero network density, simply from Services' large household consumption share and heavy within-sector input reuse; the network itself adds only an additional 9%. Resource, by contrast, sees its (much smaller) Domar weight more than double when the network is switched on. The general lesson, applicable well beyond this specific calibration: size (the Domar weight $\lambda_{D,i}$) determines which sector absorbs a generic aggregate shock; rigidity ($\hat{\delta}_i$) converts that exposure into welfare cost; and the network amplifies an already-large sector's exposure more than it creates new exposure from nothing.

6.4 Which Sector Prefers Which Regime

A related and initially counterintuitive finding, developed formally in Appendix A §A.11, is that the sector most directly exposed to the exchange rate on impact (Resource, with the largest exchange-rate pass-through coefficient $\Gamma_1 = 0.234$ versus Services' near-zero $\Gamma_3 = 0.006$) is *not* the sector that most prefers a peg. Pegging does not eliminate the need for relative-price adjustment following a foreign-price or productivity shock — it relocates that adjustment into domestic (home-goods) prices, which, by construction, are dominated by the upstream sector. This relocated-adjustment cost for Resource under a peg exceeds the direct exchange-rate cost-push it avoids, so Resource, despite having the largest Γ_i, nets out *preferring float* between the two corner regimes. Services, three links removed from the border and the stickiest sector, is doubly insulated from the peg's relocated-adjustment burden; its inflation variance falls by nearly half under a peg relative to float, and because Services' welfare weight κ_3 dwarfs the other two sectors', its own preference for the peg (conditional on the two corner regimes only) dominates the aggregate ranking between Float and Peg. Managed float dominates for *every* sector simultaneously, precisely because it damps the exchange rate's contribution to inflation variance without fully re-imposing the peg's relative-price-relocation cost — it need not pick a sectoral winner to improve on both corner regimes at once.

6.5 Cross-Country Robustness: Korea and Czechia

Table 4 repeats the headline welfare comparison using independently constructed national IO calibrations for South Korea and Czechia. The ranking — Managed float dominant, Peg worst by a wide margin — replicates in both, with both economies' losses uniformly higher than Chile's, consistent with both having denser domestic networks and higher import exposure than Chile in the underlying IO data — itself supporting evidence that the mechanism is a genuine network story rather than a Chile-specific artifact.

Table 4: Cross-country welfare loss by regime ($\times 10^{-4}$)

	Float	Managed Float	Peg
Chile	25.47	10.17	102.05
South Korea	38.91	14.43	133.81
Czechia	32.20	11.82	84.03

A separate, weaker test asks whether Services' "mostly indirect exposure via the network" finding (Section 6.3) is a general pattern across downstream, low-import sectors. Pooling all three calibrations (nine sector-level observations) gives only a moderate, noisy relationship between how much a sector sources domestically and how much of its import exposure is indirect — an honest limitation given only three sectors per calibration. What holds robustly across all three calibrations *separately*, however, is that Services sources the *least* domestically of any sector in every single calibration, yet still derives a large share of its exposure indirectly, because the small amount it does source domestically is concentrated in the single most import-heavy supplier rather than spread evenly — a composition effect, not a volume effect. A fully disaggregated, many-sector calibration (Section 8) is the natural next step to test this pattern with statistical power beyond nine observations.

7 Robustness Analysis

Because the headline welfare ranking rests, per Section 6.2, so heavily on two specific calibration choices — the risk-premium shock's standard deviation σ_{RP} and the debt-elasticity coefficient Ψ — this section subjects the model to an extensive robustness campaign: in total, roughly 350 additional Dynare solves beyond the three headline country calibrations, organized into a sequence of exercises. Throughout, all real-Chile-calibration exercises hold the sector count, Calvo frequencies, and all shock processes other than the one(s) explicitly varied fixed at their Table 1 baseline values, and network-density exercises use a continuous scaling dial $\rho \in [0, 2]$ applied to the off-diagonal entries of Ω^H ($\rho = 0$: no domestic cross-sector network, each sector uses only labor and imports; $\rho = 1$: the real Chile calibration; $\rho = 2$: double density), with the diagonal (own-sector reuse) held fixed and the labor share α_i absorbing the freed or additionally consumed cost share so that cost shares continue to sum to one at every ρ.

7.1 Network Density and the Optimal Managed-Float Response

Does the network itself matter for welfare, beyond simply being "an open economy"? Scaling network density ρ continuously on a stylized triangular network calibrated to match the real Chile economy's total off-diagonal cost-share mass, the network premium — the gap between welfare loss with the network at its real density versus none at all — roughly triples the Float/Peg welfare gap as density rises, and reallocates *which* sector bears the cost by regime. Does the *optimal* managed-float FX-response weight ϕ_s^* itself shift with network density, or is the network-density result purely a level effect? A twelve-point grid search over ϕ_s at each of three network densities ($\rho = 0, 1, 2$) shows the optimum shifts materially: $\phi_s^* = 0.15$ at $\rho = 0$, rising to 0.20 at the real Chile density, and to 0.30 at double density — confirming that denser production networks call for a *more* aggressive exchange-rate response, not just a larger welfare loss at any given response. The welfare curve is flat within roughly 2–3% of each optimum across a wide neighborhood of ϕ_s, so this is a real, monotonic shift rather than a knife-edge result, but getting ϕ_s within about ± 0.1 of correct matters far more than getting it exactly right.

7.2 Risk-Premium Volatility

Scaling the risk-premium shock's standard deviation from zero to three times its calibrated value, holding every other shock and parameter fixed, shows that with the shock switched off entirely, Peg is not dominated at all — it is fractionally *better* than Float (a ratio of 0.95). Half of the calibrated shock size already flips this to Peg being 1.7 times worse than Float; the full calibrated size gives the headline 3.6×; and three times the calibrated size gives 14.5×. This is the single most important robustness finding in the paper in the sense that it directly qualifies the headline result: the "Peg is dominated" conclusion is not an artifact of an implausible shock size (even a modest fraction of the calibrated size already produces the qualitative ranking), but its *margin* is highly sensitive to a shock size that is, honestly, an assumed-equal-to-every-other-shock convention rather than an externally estimated moment.

7.3 ψ-Sensitivity, on Its Own and Jointly with Network Density and Shock Size

The debt-elastic risk-premium coefficient Ψ (baseline 0.020) is itself a literature default with a known numerical pitfall: values too close to zero can generate a near-unit-root, knife-edge blowup in net-foreign-asset dynamics that is a linearization artifact rather than real economics. Scaling Ψ from 0.25× to 4× its baseline (0.005 to 0.080) shows the qualitative ranking (Peg worst) is robust across the entire grid, but the *margin* is not constant: the Peg/Float welfare ratio shrinks from 5.4× at the lowest Ψ to 2.7× at the highest, and the risk-premium shock's share of Peg's total loss (the 75% headline figure, at baseline $\Psi = 0.020$) ranges from 83% at the lowest Ψ down to 58% at the highest — risk-premium/UIP remains the single dominant channel throughout, but the exact 75% figure is itself a function of an under-scrutinized calibration choice.

Crossing this ψ-sensitivity check with network density (45 additional solves: five Ψ values $\times$ three densities $\times$ three regimes) shows the pattern found on the real Chile calibration generalizes essentially unchanged: the Peg/Float ratio moves from roughly 5.3–5.5$\times$ at the lowest Ψ to 2.7–2.8$\times$ at the highest, at *every* network density from zero to double — ψ-sensitivity and network density are essentially orthogonal margins of the model.

A separate, nine-point joint grid (scaling Ψ and the risk-premium shock's own volatility σ_{RP} simultaneously, Peg regime only) tests whether the two calibration choices flagged above reinforce or offset one another. A log-log regression of Peg's welfare loss on both scale factors finds an elasticity with respect to σ_{RP} of approximately $+1.44$ (dominant and convex — doubling the shock's size more than doubles the loss) against an elasticity with respect to Ψ of approximately -0.24 (real but secondary), with a non-trivial interaction term of approximately -0.16: Ψ's dampening effect on Peg's loss is *stronger*, not weaker, when the risk-premium shock itself is larger (the ratio $W(\Psi{=}2\times)/W(\Psi{=}0.5\times)$ shrinks from 0.85 at low σ_{RP} to 0.62 at high σ_{RP}). The two calibration choices flagged in Section 6.2 reinforce, rather than offset, one another.

7.4 Price Rigidity: Uniform and Services-Specific

Two complementary cuts test whether the "Why Services?" mechanism of Section 6.3 is specifically about *Services'* own rigidity, or reflects aggregate rigidity more generally.

Uniform rigidity. Scaling all three sectors' stickiness $(1-\hat{\delta}_i)$ by a common multiplier $\kappa \in \{0.5, 0.75, 1.0, 1.15\}$, crossed with network density, is bounded above by a real feasibility constraint: the current calibration's Services sector is already close enough to the Calvo-rigidity ceiling that a multiplier of 1.5 or above implies a *negative* implied reset probability, caught cleanly by the sweep script's own feasibility guard rather than by a numerical crash. Within the feasible range, Peg's welfare loss rises monotonically and steeply in aggregate rigidity at every network density (for example, at double network density: 60, 104, 182, and 251, in units of 10^{-4}, across the feasible κ grid) — rigidity compounds with Peg's existing risk-premium problem, a clean and unambiguous result. Float and Managed, by contrast, are *not* monotonic across the same grid, and the topmost feasible point ($\kappa = 1.15$) should be read as a stress test rather than a reliable point: at that point, Services' Calvo persistence index $\hat{\delta}_3$ collapses to 0.0015, an order of magnitude closer to zero than the already-small baseline value (0.031), right at the model's own numerical feasibility floor, and the welfare weight in equation (12) scales like $(1 - \hat{\delta}_i)/\hat{\delta}_i$, making this point numerically fragile by construction.

Services-only rigidity. Moving only Services' own Calvo reset probability $\delta_3 \in \{0.05, 0.10, 0.16(\text{baseline}), 0.25, 0.40, 0.60\}$, holding both other sectors fixed at their baseline calibration, produces a cleaner picture, with no near-degenerate points anywhere in the grid: Peg's welfare loss falls *monotonically* as Services becomes more flexible, at every network density (for example at the real Chile density: 122.9, 113.7, 102.1, 85.0, 60.7, and 37.0, in units of 10^{-4}, as δ_3 rises from 0.05 to 0.60) — a clean, robust confirmation that Services' own rigidity is a first-order driver of Peg's welfare dominance, not merely a coincidental correlate of aggregate stickiness. Float and Managed, in contrast, are hump-shaped rather than monotonic in this same grid, peaking around $\delta_3 \approx 0.25$–0.40: the welfare weight $(1-\hat{\delta}_i)/\hat{\delta}_i$ and the Phillips-curve slope's own sensitivity to δ_3 move in offsetting directions as rigidity falls, a genuine (if secondary) non-monotonicity rather than a numerical artifact, since this grid stays well clear of the feasibility floor identified in the uniform-rigidity cut above.

7.5 Network Topology: Beyond the Density Dial

Every exercise up to this point varies only the *density* of a single, fixed triangular network shape. This subsection asks whether the results are specific to that shape by constructing two alternative topologies at matched total off-diagonal cost-share mass (0.6501, identical to the baseline triangular network): a **hub-and-spoke** network with Manufacturing as the central node (Resource and Services each link only to Manufacturing, with no direct Resource–Services link), and a **chain** network with strictly one-directional pass-through, Resource $\rightarrow$ Manufacturing $\rightarrow$ Services, and no back-links at all.

The welfare ranking — Peg worst, Managed best — survives in *every* topology (Table 5), and the chain topology, with its most concentrated pass-through structure, is uniformly worse than the baseline triangle for every regime, with the hub-and-spoke topology intermediate. The optimal managed-float FX-response weight shifts with topology as well as with density: $\phi_s^* = 0.20$ for the triangle, but 0.30 for both the hub-and-spoke and chain topologies, consistent with (and reinforcing) the density-based finding of Section 7.2's companion exercise above. Recomputing the analytical import-centrality decomposition of Section 6.3 for each topology shows Services' indirect-exposure share is not an artifact of the triangular shape — if anything it strengthens under more concentrated pass-through structures, rising from 41% (triangle) to 46% (hub-and-spoke) to 63% (chain) of Services' total import centrality.

Table 5: Welfare loss by network topology and regime ($\times 10^{-4}$)

Topology	Float	Managed	Peg
Triangle (baseline)	25.5	10.2	102.1
Hub-and-spoke (Manufacturing hub)	28.9	11.3	113.5
Chain (Resource→Manuf.→Services)	33.7	12.7	115.5

7.6 Regime-Rule Variants

The baseline float regime targets Rubbo's divine-coincidence index with conventional (non-aggressive) Taylor coefficients $(\phi_\pi, \phi_y) = (1.5, 0.5)$. This subsection asks two further questions: does targeting a different price index (strict domestic/PPI or strict consumer-price/CPI inflation, instead of the divine-coincidence index) matter, and does the Taylor rule's own aggressiveness matter?

At the baseline, non-aggressive Taylor coefficients, adding two new policy-rule variants — strict CPI-inflation targeting and strict PPI (domestic-bundle)-inflation targeting, in place of the divine-coincidence index — to the model (verified, via a regression check re-running the three original regimes through the edited model files, to reproduce the original headline numbers exactly, confirming the new branches are additive and non-breaking) shows both new variants land statistically indistinguishable from Float: 25.36 (CPI) and 25.26 (PPI) versus Float's 25.47 ($\times 10^{-4}$), all within about one percent of one another. **Targeting the theoretically privileged divine-coincidence index, at these conventional policy coefficients, buys almost nothing over naively targeting CPI or PPI inflation**; essentially all of Managed float's $\approx 2.5\times$ improvement comes from its explicit exchange-rate response term, not from smarter inflation targeting.

This picture changes once the Taylor rule is made aggressive. Setting $\phi_\pi = 5$ (a numerical proxy for a very aggressive, near-strict inflation-targeting rule) and $\phi_y = 0$, at the real Chile network density, gives a markedly different ranking among the three price-index targets: strict PPI targeting (10.55, $\times 10^{-4}$) outperforms strict divine-coincidence targeting (12.57), which in turn outperforms strict CPI targeting (18.48). **Once policy is aggressive enough, targeting the divine-coincidence index is no longer the best choice among simple alternatives; targeting domestic (PPI) inflation strictly is**, reversing the conventional-coefficient finding above. A companion coarse grid search over $(\phi_\pi, \phi_y) \in \{1.5, 3, 5\} \times \{0, 0.5, 1\}$ on the divine-coincidence-index (float) rule finds its best point in this 3×3 grid, $(\phi_\pi, \phi_y) = (5, 0.5)$, gives a welfare loss of 9.55 ($\times 10^{-4}$) — essentially matching, and fractionally beating, Managed float's headline 10.17, *without any explicit exchange-rate response term at all.* I flag this finding explicitly as preliminary rather than settled: it is a coarse 3×3 grid on the float branch only, and Managed float's own (ϕ_π, ϕ_y) coefficients were left at their original, non-aggressive baseline rather than re-optimized at the same level of aggressiveness, so the comparison is not yet apples-to-apples. What it does establish cleanly is that the baseline $\phi_\pi = 1.5$ materially undersells how well plain divine-coincidence-index targeting can do; ϕ_π itself had never previously been swept in this project.

7.7 The Network Premium on the Real Calibration

Every network-density exercise above uses a stylized triangular network to permit continuous scaling. As a direct check that the network channel matters on the paper's *actual* headline calibration — not merely on an illustrative proxy — I construct a no-network counterfactual by zeroing all nine entries of Chile's real, dense Ω^H matrix (including the diagonal, so each sector uses only labor and imports), letting the labor share α_i absorb the freed cost share, re-deriving the non-stochastic steady state, and re-simulating all three regimes. The resulting network premium is substantial and consistent across regimes: welfare loss with the real network in place, relative to the no-network counterfactual, is 213% higher for Float (25.47 versus 8.13), 169% higher for Peg (102.05 versus 37.94), and 92% higher for Managed (10.17 versus 5.30). The production network itself — not simply the fact of being an open economy — drives the majority of the welfare loss in every regime on the paper's real, headline calibration, and amplifies Float's loss proportionally more than Managed's.

7.8 Sector-Specific Export Demand

The baseline model allocates aggregate exports across sectors via household consumption shares $\boldsymbol{\beta}^H$ (heavily weighted toward Services), rather than via each sector's actual, empirically very different export intensity (heavily weighted toward Resource: 60% of Resource's own output is exported, versus under 4% for Services). This understates the model's ability to represent "the export-heavy sector is hit by a terms-of-trade shock; how does the rest of the economy feel it through the network," since under the baseline allocation the shock always spreads via consumption shares rather than genuine export exposure. Replacing the single aggregate export equation with three sector-specific export equations, recalibrated (via a damped fixed-point iteration on each sector's export-scale constant) so that steady-state export intensity matches the real, empirical export shares while keeping the aggregate trade balance and net-foreign-asset target unchanged, and re-deriving the steady state accordingly, shows the export-price (terms-of-trade) shock's contribution to Peg's welfare loss *falls* substantially once exports are allocated realistically: from 2.16 to 1.17 ($\times 10^{-4}$), a 46% reduction, because the baseline consumption-share allocation was overstating how much export-price risk the (Services-heavy) consumption-share proxy exposes Peg to, relative to the true (Resource-heavy) export exposure. Peg's total welfare loss falls correspondingly, from 102.05 to 90.45 ($\times 10^{-4}$), an 11.4% reduction, though the qualitative ranking is entirely unaffected. A companion network-premium check under this more realistic export allocation reproduces the Section 7.7 finding closely (Float +190%, Peg +169%, Managed +88%), and additional sensitivity sweeps over the export-demand elasticity and the reallocation intensity both show smooth, monotonic behavior with no ranking flips.

7.9 Second-Order (Nonlinear) Welfare Check

All results above use the model's order-1 (log-linear) solution, evaluating the (already second-order) welfare functional (12) at first-order-accurate equilibrium paths — the standard linear-quadratic shortcut, valid whenever the first-order solution is certainty-equivalent. This equivalence can, in principle, fail in this specific model because of two open-economy features absent from Rubbo's closed economy: the debt-elastic UIP wedge is nonlinear in the net foreign asset position, and the net foreign asset position itself has a root close to the inverse discount factor, so precautionary (Jensen's-inequality) terms need not average out over any short horizon. Re-solving the model to a genuine order-2 (pruned) approximation and simulating 260,000 periods (rather than relying on Dynare's analytic higher-order moments, which require stationarity the model's own price-level and exchange-rate-level state variables do not satisfy under an inflation-targeting rule), computing $\mathbb{E}[X^2]$ directly from the simulated path rather than from the sample variance, and comparing against an identical-seed, identical-length order-1 simulation to control for Monte Carlo noise, shows the ranking is fully second-order-robust: every regime's loss rises by at most 2% (Float +2.0%, Managed +0.6%, Peg +0.7%), nowhere near enough to threaten Managed < Float < Peg. Decomposing the correction further shows the risk-adjusted mean shifts implied by the nonlinear model are real and economically interpretable — Peg's risk-adjusted output gap sits roughly ten times larger, in absolute value, than Float's or Managed's, and net foreign assets carry a larger precautionary buffer exactly where the UIP/risk-premium channel is least offset by exchange-rate movement — but because the welfare loss is quadratic, squaring these small mean shifts contributes less than half a percent of the total squared loss in every regime; essentially all of the modest second-order correction instead reflects a small, common variance amplification across all three regimes, not a risk-adjusted-steady-state effect specific to any one of them.

7.10 Summary of the Robustness Campaign

Table 6 collects the qualitative conclusion of every exercise in this section. The headline ranking Managed < Float $\ll$ Peg is, without a single exception across roughly 350 additional model solves spanning network density, network topology, uniform and sector-specific price rigidity, the risk-premium shock's size and closing-device elasticity (on their own and jointly, and jointly with network density), and a range of alternative policy-rule designs, qualitatively robust. What is *not* robust to the same degree is the precise *margin* of Peg's dominance, which is highly sensitive to the risk-premium shock's calibrated volatility and its interaction with the debt-elasticity coefficient Ψ — exactly the two parameters this paper's underlying financial-account machinery introduces relative to Rubbo's closed economy and to Qiu, Wang, Xu and Zanetti's (2026) static open economy.

Table 6: Robustness campaign summary: does Managed $<$ Float $\ll$ Peg survive?

Exercise	Ranking survives?	Key additional finding
Network density ($\rho \in [0, 2]$)	Yes	Network premium roughly triples Float/Peg gap
Optimal ϕ_s vs. density	—	ϕ_s^*: 0.15 $\rightarrow$ 0.20 $\rightarrow$ 0.30 as ρ rises
Risk-premium volatility	Yes (mostly)	Off entirely: Peg $\approx$ tied with Float
ψ sensitivity	Yes	Peg/Float ratio: 5.4$\times$ $\rightarrow$ 2.7$\times$ across grid
$\psi\times$ network density	Yes	Interaction is negligible
$\psi \times \sigma_{RP}$ joint	Yes	Genuine reinforcing interaction (≈ -0.16)
Uniform rigidity $\times$ density	Yes	Peg monotonic; Float/Managed not
Services-only rigidity $\times$ density	Yes	Peg monotonic; Float/Managed hump-shaped
Network topology	Yes	Chain worst; ϕ_s^* shifts with shape too
Strict CPI/PPI-IT (baseline coeff.)	Yes	All three $\approx$ indistinguishable from Float
Strict CPI/PPI-IT (aggressive coeff.)	Yes	PPI $<$ DC-index $<$ CPI (reversed!)
Dual-mandate (ϕ_π, ϕ_y) grid	Yes (caveated)	Aggressive plain float may rival Managed
Network premium (real calibration)	—	+213%/+169%/+92% (Float/Peg/Managed)
Sector-specific exports	Yes	Peg's loss falls 11.4% under realistic exports
Second-order (nonlinear) welfare	Yes	Losses rise $\leq$2% in every regime
Cross-country (Korea, Czechia)	Yes	Both losses uniformly higher than Chile's

8 Discussion

8.1 Policy Implications

Three practical implications follow from this paper's results, with appropriate caveats attached to each. First, for a small open economy with a production network resembling this paper's calibrations, a managed float with a modest, positive response to the exchange rate is likely to dominate both corner regimes by a wide margin, and this conclusion appears robust to essentially every dimension of model misspecification examined in Section 7. Second, the specific mechanism generating this ranking — a financial risk-premium/UIP channel, not the terms-of-trade channel emphasized by the classical literature — means that policies aimed at reducing exposure to the risk-premium channel specifically (for example, reserve accumulation along the lines of Fanelli and Straub, 2021, or policies reducing the debt-elasticity of the country's risk premium) may be a more direct lever on welfare than exchange-rate policy design per se, a possibility this paper's model is not currently equipped to evaluate directly since Ψ and σ_{RP} are treated as exogenous primitives rather than policy-influenced objects. Third, Section 7.6's finding that policy aggressiveness, not merely the choice of price index, is a first-order determinant of welfare — and that the theoretically privileged divine-coincidence index is not obviously the best practical target once policy is aggressive — suggests that calibrating how hard a central bank *can* lean against inflation, given its other constraints, may matter as much as getting the target index exactly right.

8.2 Limitations and Directions for Future Work

Honestly assessed, this paper is not yet ready for submission to a top-five journal, and treating the current draft as a checkpoint rather than a near-final product is the right frame. Several concrete gaps remain. First and most importantly, the three-sector aggregation used throughout, while sufficient to establish the qualitative mechanism and to permit the extensive robustness campaign of Section 7, is coarser than a competing paper in this space (Qiu, Wang, Xu and Zanetti, 2026) that has already cleared peer review using a considerably more disaggregated cross-border input–output dataset (the World Input-Output Database, 43 countries by 56 sectors); a genuinely multi-sector calibration using an analogous disaggregated database (for example, OECD's Trade in

Value Added tables) is the most important remaining step, and is likely to be requested by referees. Second, and directly related to the robustness campaign's central finding, several parameters driving the headline result — most importantly the risk-premium shock's standard deviation and the debt-elasticity coefficient Ψ, but also the within- and across-sector elasticities of substitution and the shock persistence parameters — are literature defaults rather than estimates specific to this setting or this country; an independently estimated risk-premium process, in particular, is a necessary input before the margin (though not, per Section 7, the qualitative ranking) of the peg-dominated result can be treated as more than suggestive. Third, the paper currently contains no empirical validation section tying the model's quantitative magnitudes to data from an actual historical exchange-rate regime episode; Silva's (2024) Chilean CPI-elasticity application is a natural template for such an exercise, and a natural complement rather than a substitute for this paper's own normative, regime-comparison exercise. Fourth, even a complete, referee-ready draft addressing the above typically undergoes two to three referee rounds over one to two or more years at the journals this paper aspires to; that clock has not yet started. The necessary condition for a genuine contribution — the paper fills a gap explicitly flagged as open by the authors of the closest competing paper themselves (Section 2) — appears to be met; the sufficient condition, a complete and fully referee-proofed paper, remains realistically several months to a year of further work away, beginning with the multi-sector calibration and a fuller engagement with Qiu, Wang, Xu and Zanetti's (2026) output-gap-weight derivation directly in the model section rather than solely as a literature-review citation.

Beyond these gaps required for submission readiness, several extensions raised during the robustness campaign (Section 7) but flagged there as preliminary would repay further work: a formally re-optimized managed-float rule at the same level of policy aggressiveness as the dual-mandate grid search of Section 7.6, to settle whether an aggressive plain float rule genuinely rivals (rather than merely appears to rival) the managed float once both are given the same optimization effort; and a dominant-currency-pricing extension along the lines of Gopinath and Itskhoki (2022), replacing this paper's producer-currency-pricing assumption with a mixed invoicing regime, which would alter the exchange-rate pass-through vector Γ in ways this paper's current model cannot represent.

9 Conclusion

This paper extends Rubbo's (2024) closed-economy theory of production networks and monetary policy to a small open economy, and shows that her central divine-coincidence result — the existence of a single inflation index whose targeting simultaneously closes the output gap — generically breaks down once the exchange rate is introduced as a second, independent cost-push channel alongside sectoral productivity. Quantitatively, calibrating the model to three real national input–output datasets, I find that a managed float dominates both a free float and a hard peg, and that this ranking is driven overwhelmingly by a financial risk-premium/uncovered-interest-parity channel rather than the terms-of-trade mechanism emphasized by the classical open-economy literature. An extensive robustness campaign — spanning network density and shape, price rigidity, the calibration of the risk-premium process, and alternative monetary policy rules — confirms this qualitative ranking is remarkably stable, while surfacing two further, more nuanced findings about policy aggressiveness and price-index choice that merit further, more careful investigation before being treated as settled. The paper's relationship to the contemporaneous and independent work of Qiu, Wang, Xu and Zanetti (2026) is, I have argued, one of corroboration and natural extension rather than redundancy: their own stated top-priority direction for future research — relaxing the assumption of financial autarky — is precisely the machinery this paper's contribution is built around, and precisely the channel this paper's quantitative analysis identifies as the true driver of the exchange-rate-regime welfare ranking.

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A Proofs

This appendix collects detailed, step-by-step proofs of every lemma and proposition stated in Section 4. The closed-economy results (Lemmas 1–A.4 and Propositions 1–2) reproduce Rubbo's (2024) own derivations in full, restated here for completeness and because the open-economy results (§A.8) are obtained by an essentially identical argument applied to a modified marginal cost equation.

A.1 Notation

Lower-case letters denote log-deviations from a zero-inflation, zero-markup non-stochastic steady state throughout. $\boldsymbol{\alpha} \in \mathbb{R}^N$ is the labor cost-share vector, $\Omega^H \in \mathbb{R}^{N\times N}$ the domestic input–output matrix, $\boldsymbol{\beta}^H \in \mathbb{R}^N$ household consumption shares across domestic sectors, $\Delta = \mathrm{diag}(\hat{\delta}_1, \ldots, \hat{\delta}_N)$ the effective Calvo persistence matrix, and $(I - \Omega^H)^{-1}$, $(I - \Omega^H\Delta)^{-1}$ the (rigidity-unadjusted and rigidity-adjusted) Leontief inverses. $\boldsymbol{\lambda}^\top = (\boldsymbol{\beta}^H)^\top (I - \Omega^H)^{-1}$ are Domar weights.

A.2 Log-Linearized Building Blocks

Marginal cost. Since production is constant-returns-to-scale and productivity is Hicks-neutral, the unit cost function is $MC_{it} = \tilde{c}_i(W_t, \{P_{jt}\})/A_{it}$. By Shephard's lemma, $\partial \log MC_{it}/\partial \log W_t = \alpha_i$ (labor's cost share) and $\partial \log MC_{it}/\partial \log P_{jt} = \Omega^H_{ij}$ (input j's cost share); and $\partial \log MC_{it}/\partial \log A_{it} = -1$. Log-linearizing and stacking across sectors,

$$\boldsymbol{mc}_t = \boldsymbol{\alpha}\, w_t + \Omega^H \boldsymbol{p}_t - \log \mathbf{A}_t. \tag{13}$$

Price-level dynamics and Calvo reset price. With a fraction δ_i of sector-i firms resetting price each period, log-linearizing $P_{it}^{1-\varepsilon_i} = (1-\delta_i)P_{it-1}^{1-\varepsilon_i} + \delta_i (P_{it}^*)^{1-\varepsilon_i}$ around the zero-inflation steady state gives $p_{it}^* - p_{it-1} = \pi_{it}/\delta_i$. Log-linearizing the optimal Calvo reset price around the same steady state gives $p_{it}^* = (1 - \rho(1-\delta_i)) \sum_{s\geq 0} [\rho(1-\delta_i)]^s \mathbb{E}_t mc_{it+s}$. Combining these two and solving forward one step yields the sectoral Phillips curve

$$\pi_{it} = \hat{\delta}_i (mc_{it} - p_{it-1}) + \rho(1-\hat{\delta}_i)\mathbb{E}_t \pi_{it+1}, \qquad \hat{\delta}_i \equiv \frac{\delta_i(1-\rho(1-\delta_i))}{1-\rho\delta_i(1-\delta_i)}, \tag{14}$$

or, in matrix form, $\boldsymbol{\pi}_t = \Delta(\boldsymbol{mc}_t - \boldsymbol{p}_{t-1}) + \rho(I-\Delta)\mathbb{E}\boldsymbol{\pi}_{t+1}$.

Real wage. Log-linearizing labor supply $W_t/P_t^C = C_t^\gamma L_t^\varphi$ and using goods market clearing $c_t \approx y_t$ together with Lemma 1 below,

$$w_t - (\boldsymbol{\beta}^H)^\top \boldsymbol{p}_t = (\gamma + \varphi)\tilde{y}_t + \boldsymbol{\lambda}^\top \log \mathbf{A}_t. \tag{15}$$

A.3 Proof of Lemma 1 (Natural Output)

The flexible-price equilibrium is efficient, so it solves the planner's problem max $U(\mathcal{C}(\{Y_i - \sum_j X_{ji}\}), L)$ subject to $Y_i = A_i F_i(\{X_{ij}\}, L_i)$. Decompose this into an inner problem — given total labor L, allocate it (and produce) efficiently across sectors, defining $C^*(L; \mathbf{A})$ — and an outer problem choosing L. By constant returns to scale of every F_i and of the consumption aggregator $\mathcal{C}$, C^* is homogeneous of degree one in L. The outer first-order condition is $C^{*-\gamma}\partial C^*/\partial L = L^\varphi$. By the envelope theorem applied to the inner Lagrangian, $\partial C^*/\partial L = \nu_L$ (the shadow value of labor) and $\partial C^*/\partial A_i = \nu_i\, \partial(A_i F_i)/\partial A_i = \nu_i Y_i^*$ (the shadow value of good i times its optimal output). Converting to logs and using that, in competitive equilibrium, $\nu_i = P_i/(P^C C^{*\gamma})$, gives $\partial \log C^*/\partial \log A_i = P_i Y_i^*/(P^C C^*) \equiv \lambda_i$, the Domar weight. Because C^* is homogeneous of degree one in L, Euler's theorem gives $C^* = L\nu_L$; combining this with the outer first-order condition and the labor-supply and labor-demand conditions in the natural equilibrium (using $y_{\text{nat}} - l_{\text{nat}} = \boldsymbol{\lambda}^\top \log \mathbf{A}_t$, itself following from Step 4's Domar-weight identity applied to the aggregate CRS production function) yields, after collecting terms, $y_{t,\text{nat}} = \frac{1+\varphi}{\gamma+\varphi}\boldsymbol{\lambda}^\top \log \mathbf{A}_t$. ■

A.4 Proof of Lemma (Output Gap Equals Employment Gap)

By constant returns to scale, aggregate output in both the actual and the natural equilibrium is given by the same inner value function, $y_t = C_t^*(L_t; \mathbf{A}_t)$, $y_{t,\text{nat}} = C_t^*(L_{t,\text{nat}}; \mathbf{A}_t)$. Log-linearizing around $L_{t,\text{nat}}$ and applying the envelope theorem exactly as in the proof of Lemma 1, $\partial \log C^*/\partial \log L = L\nu_L/C^*$, which equals one at the natural equilibrium because $\nu_L = C^{*\gamma}L^\varphi$ and the outer first-order condition holds with equality. Hence $\tilde{y}_t \equiv y_t - y_{t,\text{nat}} = 1 \cdot (l_t - l_{t,\text{nat}}) \equiv \tilde{l}_t$. ■

A.5 Proof of Lemma (Output Gap Equals Minus the Sales-Weighted Markup)

From the household's labor-supply condition, in both the actual and natural equilibria, $(w_t - p_t^C) - (w_{t,\text{nat}} - p_{t,\text{nat}}^C) = (\gamma + \varphi)\tilde{y}_t$, using $\tilde{y}_t = \tilde{l}_t$ from the preceding lemma. Writing the price gap $\tilde{\boldsymbol{p}}_t \equiv \boldsymbol{p}_t - \boldsymbol{p}_{t,\text{nat}}$, the marginal-cost gap satisfies $\tilde{\boldsymbol{mc}}_t = \boldsymbol{\alpha}\tilde{w}_t + \Omega^H \tilde{\boldsymbol{p}}_t$ (subtracting equation (13) evaluated at the natural equilibrium, where $\log \mathbf{A}_t$ cancels). Since markups are $\mu_{it} = p_{it} - mc_{it}$ and are zero in the natural equilibrium, $\tilde{\boldsymbol{p}}_t = \tilde{\boldsymbol{mc}}_t + \boldsymbol{\mu}_t = \boldsymbol{\alpha}\tilde{w}_t + \Omega^H \tilde{\boldsymbol{p}}_t + \boldsymbol{\mu}_t$, so $(I - \Omega^H)\tilde{\boldsymbol{p}}_t = \boldsymbol{\alpha}\tilde{w}_t + \boldsymbol{\mu}_t$, and pre-multiplying by $(I - \Omega^H)^{-1}$ and then by $(\boldsymbol{\beta}^H)^\top$ gives $(\boldsymbol{\beta}^H)^\top \tilde{\boldsymbol{p}}_t = (\boldsymbol{\lambda}^\top \boldsymbol{\alpha})\tilde{w}_t + \boldsymbol{\lambda}^\top \boldsymbol{\mu}_t = \tilde{w}_t + \boldsymbol{\lambda}^\top \boldsymbol{\mu}_t$, using the constant-returns-to-scale identity $\boldsymbol{\lambda}^\top \boldsymbol{\alpha} = 1$ (equivalently $(I - \Omega^H)^{-1}\boldsymbol{\alpha} = \mathbf{1}$, since $\alpha_i + \sum_j \Omega_{ij}^H = 1$ for every i). Substituting into the labor-supply-gap equation above, $\tilde{w}_t - (\tilde{w}_t + \boldsymbol{\lambda}^\top \boldsymbol{\mu}_t) = (\gamma + \varphi)\tilde{y}_t$, i.e. $(\gamma + \varphi)\tilde{y}_t = -\boldsymbol{\lambda}^\top \boldsymbol{\mu}_t$. ■

A.6 Proof of Proposition 1 (Sectoral Phillips Curve)

Substituting equation (13) into (14) and using $\boldsymbol{p}_t = \boldsymbol{p}_{t-1} + \boldsymbol{\pi}_t$ gives, after collecting all $\boldsymbol{\pi}_t$ terms on the left, $(I - \Delta\Omega^H - \Delta\boldsymbol{\alpha}(\boldsymbol{\beta}^H)^\top)\boldsymbol{\pi}_t = \Delta\boldsymbol{\alpha}(\gamma + \varphi)\tilde{y}_t + \Delta(\boldsymbol{\alpha}\boldsymbol{\lambda}^\top - I)\log \mathbf{A}_t + \Delta(\boldsymbol{\alpha}(\boldsymbol{\beta}^H)^\top - I + \Omega^H)\boldsymbol{p}_{t-1} + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1}$, where the real-wage equation (15) has been substituted for w_t. The left-hand matrix factors as $(I - \Delta\Omega^H)(I - \Delta(I - \Omega^H\Delta)^{-1}\boldsymbol{\alpha}(\boldsymbol{\beta}^H)^\top)$, using the identity $(I - \Delta\Omega^H)^{-1}\Delta = \Delta(I - \Omega^H\Delta)^{-1}$ (verified by direct multiplication). The second factor is a rank-one perturbation of the identity, invertible by the Sherman–Morrison formula, $(I - \mathbf{u}\mathbf{v}^\top)^{-1} = I + \mathbf{u}\mathbf{v}^\top/(1 - \mathbf{v}^\top\mathbf{u})$, with $\mathbf{u} = \Delta(I - \Omega^H\Delta)^{-1}\boldsymbol{\alpha} \equiv \tilde{A}\boldsymbol{\alpha}$ and $\mathbf{v} = \boldsymbol{\beta}^H$. Defining $\kappa \equiv 1 - (\boldsymbol{\beta}^H)^\top\tilde{A}\boldsymbol{\alpha}$, direct computation of $(I - \Delta\Omega^H - \Delta\boldsymbol{\alpha}(\boldsymbol{\beta}^H)^\top)^{-1}\Delta\boldsymbol{\alpha}$ using this inverse gives exactly $\tilde{A}\boldsymbol{\alpha}/\kappa \equiv \mathcal{B}$, the output-gap coefficient in equation (8). Carrying the analogous computation through for the productivity and lagged-price terms, and collecting them into $-\mathcal{V}\boldsymbol{\chi}_t$ with $\boldsymbol{\chi}_t \equiv (I - \Omega^H)^{-1}\log \mathbf{A}_t - \boldsymbol{p}_{t-1}$, yields equation (8). That $\mathcal{V}\mathbf{1} = \mathbf{0}$ follows because $\mathcal{V} = (A - I)(I - \Omega^H)$ for a matrix A satisfying $A\mathbf{1} = \mathbf{1}$ (verified directly from A's definition using $(I - \Omega^H)^{-1}\boldsymbol{\alpha} = \mathbf{1}$), so $\mathcal{V}\mathbf{1} = (A - I)(I - \Omega^H)\mathbf{1} = (A - I)\boldsymbol{\alpha} = A\boldsymbol{\alpha} - \boldsymbol{\alpha}$, which vanishes because $A\boldsymbol{\alpha} = \boldsymbol{\alpha}$ by the same identity applied componentwise. ■

A.7 Proof of Proposition 2 (Divine Coincidence, Closed Economy)

From equation (14) and the markup definition $\mu_{it} = p_{it} - mc_{it}$, substituting $mc_{it} - p_{it-1} = (\pi_{it} - \rho(1-\hat{\delta}_i)\mathbb{E}_t\pi_{it+1})/\hat{\delta}_i$ gives the exact markup-inflation link $\mu_{it} = -\frac{1-\hat{\delta}_i}{\hat{\delta}_i}(\pi_{it} - \rho\mathbb{E}_t\pi_{it+1})$. Substituting this into $(\gamma+\varphi)\tilde{y}_t = -\boldsymbol{\lambda}^\top\boldsymbol{\mu}_t$ (the preceding lemma) gives $(\gamma+\varphi)\tilde{y}_t = \big(\sum_i \lambda_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}\big)(\pi_t^{DC} - \rho\mathbb{E}\pi_{t+1}^{DC})$ where π_t^{DC} is exactly the Domar-rigidity-weighted average defined in the proposition; rearranging gives the cost-push-free Phillips curve (9) directly, with no productivity term surviving because only Lemma 1–A.4's output-gap/markup link and the Calvo markup-inflation identity were used, neither of which involves $\log \mathbf{A}_t$ directly.

For uniqueness: an index $\boldsymbol{\phi}^\top\boldsymbol{\pi}_t$ has no cost-push term if and only if $\boldsymbol{\phi}^\top\mathcal{V} = \mathbf{0}^\top$, i.e. $\boldsymbol{\phi}$ lies in $\mathcal{V}$'s left null space. Since $\mathcal{V}\mathbf{1} = \mathbf{0}$ (established above), $\mathcal{V}$ has rank at most $N-1$, so its left null space is generically exactly one-dimensional, and direct verification shows $\boldsymbol{\phi} \propto \boldsymbol{\lambda}^\top(I-\Delta)\Delta^{-1}$ (component-wise, $\phi_i \propto \lambda_i(1-\hat{\delta}_i)/\hat{\delta}_i$) satisfies $\boldsymbol{\phi}^\top\mathcal{V} = \mathbf{0}^\top$. The divine-coincidence weights are therefore the unique (up to normalization) solution. ■

A.8 Proof of Proposition 3 (Sectoral Phillips Curve, Open Economy)

Replace equation (13) with the open-economy marginal cost (5), $\boldsymbol{mc}_t = \boldsymbol{\alpha}w_t + \Omega^H\boldsymbol{p}_t + \boldsymbol{\Omega}^F(p_t^* + e_t) - \log \mathbf{A}_t$. Every step of the proof of Proposition 1 above goes through verbatim with this modified marginal cost, since none of those steps used any property of $-\log \mathbf{A}_t$ beyond its being an additive, sector-specific term entering marginal cost linearly — exactly the property the new term $\boldsymbol{\Omega}^F(p_t^* + e_t)$ shares. The new term is therefore pre-multiplied, in the final collected equation, by exactly the same matrix, $(I - \Delta\Omega^H - \Delta\boldsymbol{\alpha}(\boldsymbol{\beta}^H)^\top)^{-1}\Delta$, that pre-multiplies the output-gap coefficient, which was shown above to equal $\tilde{A}/\kappa$. Since p_t^* is exogenous (foreign, small-open-economy assumption) and only e_t is a domestic state variable, this delivers exactly the new term $\Gamma\Delta e_t$ in equation (10), with $\Gamma = \tilde{A}\boldsymbol{\Omega}^F/\kappa$ as claimed, and every other term in equation (10) unchanged from the closed economy. ■

A.9 Proof of Proposition 4 (Generic Breakdown of the Divine Coincidence)

Pre-multiplying equation (10) by a candidate weighting vector $\boldsymbol{\phi}^\top$ gives $\pi_t^{\boldsymbol{\phi}} = \rho\boldsymbol{\phi}^\top(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \boldsymbol{\phi}^\top\mathcal{B}(\gamma+\varphi)\tilde{y}_t - \boldsymbol{\phi}^\top\mathcal{V}\boldsymbol{\chi}_t + (\boldsymbol{\phi}^\top\Gamma)\Delta e_t$; this has no cost-push term of either kind if and only if both $\boldsymbol{\phi}^\top\mathcal{V} = \mathbf{0}^\top$ *and* $\boldsymbol{\phi}^\top\Gamma = 0$ hold. By Proposition 2, the first condition alone already pins $\boldsymbol{\phi}$ down uniquely, up to scale, as the closed-economy divine-coincidence weight vector $\boldsymbol{\phi}^{DC}$ — there is no remaining freedom in $\boldsymbol{\phi}$ once the first condition is imposed. The second condition, $(\boldsymbol{\phi}^{DC})^\top\Gamma = 0$, is then a single additional scalar equation with zero remaining free parameters to satisfy it: generically (for almost every choice of the primitives $\Omega^H, \boldsymbol{\alpha}, \boldsymbol{\beta}^H, \Delta, \boldsymbol{\Omega}^F$, in the sense of Lebesgue measure on the space of feasible primitives), this equation fails. It holds only in the non-generic case where Γ is itself proportional to $\mathcal{B}$ (equivalently, since $\mathcal{B} \propto \tilde{A}\boldsymbol{\alpha}$ and $\Gamma \propto \tilde{A}\boldsymbol{\Omega}^F$, the case $\boldsymbol{\Omega}^F \propto \boldsymbol{\alpha}$: import exposure exactly proportional to labor intensity in every sector), since in that knife-edge case $\boldsymbol{\phi}^{DC\top}\Gamma \propto \boldsymbol{\phi}^{DC\top}\mathcal{B}$, and it can be verified directly from the definitions that $\boldsymbol{\phi}^{DC\top}\mathcal{B}$ need not itself vanish, but the *output-gap* coefficient in the DC Phillips curve absorbs the entire exchange-rate term identically to how it already absorbs the output-gap term, restoring a (modified) divine coincidence with a shifted natural-rate object rather than eliminating the exchange-rate channel outright. ■

A.10 Proof of Proposition 5 (Welfare)

To second order, the representative household's utility loss relative to the efficient allocation decomposes additively into: (i) an output-gap term, from the second-order expansion of $U(C_t, L_t)$ around the efficient point using $\tilde{y}_t = \tilde{l}_t$ (the closed-economy Lemma above), giving $-\frac{\gamma+\varphi}{2}U_C^* C^* \tilde{y}_t^2$; (ii) a within-sector term, from the fact that under Calvo pricing the cross-sectional variance of log prices within sector i is $\mathrm{Var}_f[\log P_{if,t}] \approx \frac{1-\hat{\delta}_i}{\hat{\delta}_i}\pi_{it}^2$, which generates a first-order-in-variance TFP loss of $-\frac{1}{2}\varepsilon_i \mathrm{Var}_f[\log P_{if,t}]$ in sector i, summed with Domar weights $\lambda_{D,i}$; and (iii) cross-sector terms, from the second-order Baqaee–Farhi (2019) substitution operators Φ_C, Φ_s evaluated at the model's consumption and production elasticities of substitution, capturing the welfare cost of relative-price distortions across sectors (as distinct from within-sector price dispersion). In the open economy, the marginal cost equation (5) introduces an additional argument, $e_t\boldsymbol{\Omega}^F$, into these same cross-sector operators, since different sectors face different import cost shocks and hence different relative-price distortions even absent any within-sector dispersion; no new operators are needed, only a new argument to the existing ones. ■

Remark 2 (Closed form under Cobb–Douglas cross-sector nests)**.** *This model's cross-sector aggregators are Cobb–Douglas everywhere except the outer home/foreign consumption nest: household consumption across domestic sectors, and each firm's intermediate-input bundle across domestic sectors and imports, are both Cobb–Douglas by the assumed functional form of Y_{ift}, and Cobb–Douglas delivers an Allen–Uzawa elasticity of substitution of exactly one between any pair of its arguments, independent of the exponent (cost-share) weights. With a constant elasticity σ across all pairs, the algebraic identity $\sum_{k,h} w_k w_h (X_k - X_h)(Y_k - Y_h) = 2\big[\sum_k w_k X_k Y_k - (\sum_k w_k X_k)(\sum_h w_h Y_h)\big]$ (for weights summing to one) collapses $\Phi_C(\boldsymbol{\mu}, \boldsymbol{\mu})$ to a weighted variance of the markup-gap vector, with the import composite's markup identically zero under the law of one price. Only the outer CES consumption nest ($\eta = 1.5$) contributes a genuinely non-unitary elasticity, entering as a practical stand-in for the true nested cross-elasticity between domestic and imported goods. Numerically, in the Chile calibration, this cross-sector term adds 3.3% (Float), 8.4% (Managed), and 4.5% (Peg) on top of the output-gap-plus- within-sector total, without affecting the regime ranking.*

A.11 Which Sectors Prefer Which Regime: Supporting Derivation

Two calibrated objects jointly determine each sector's regime preference. The first, direct exchange-rate exposure, is $\Gamma_i = [\tilde{A}\boldsymbol{\Omega}^F]_i/\kappa$ from equation (11): large when sector i imports directly or buys heavily from suppliers who do, discounted by how quickly prices reset along the chain. The second, the welfare weight from equation (12), is $\kappa_i \equiv \lambda_{D,i}\varepsilon_i(1 - \hat{\delta}_i)/\hat{\delta}_i$: large when sector i is a large, central domestic supplier and very sticky. In a triangular network where the upstream sector is also the direct importer (as calibrated here, $\boldsymbol{\Omega}^F$ decreasing downstream, $\boldsymbol{\lambda}_D$ increasing downstream), these two objects move in *opposite* directions across sectors: $\Gamma_1 = 0.234 \gg \Gamma_3 = 0.006$, but $\kappa_1 = 0.43 \ll \kappa_3 = 74.6$. The sector most exposed to the exchange rate on impact is the one welfare cares about least per unit of its inflation variance, and vice versa; neither parameter alone predicts the aggregate regime ranking, but their product, combined with how each regime redistributes *where* the necessary relative-price adjustment occurs, does. Applying κ_i to the simulated regime-by-regime inflation variances gives each sector's realized loss $\kappa_i \mathrm{Var}^R(\pi_i)$: Sector 1's loss is *higher* under peg (0.59) than float (0.44), despite the Γ-channel switching off entirely under a peg, because pegging relocates the adjustment burden into home-goods prices, which sector 1 dominates; Sector 3's loss falls by nearly half under peg (16.96 under float versus 8.86 under peg) because it is both insulated from direct exchange-rate exposure and largely spared the peg's relocated-adjustment burden. Since $\kappa_3 \gg \kappa_1, \kappa_2$, Sector 3's preference for peg over float dominates the aggregate ranking between the two corner regimes, even though Sector 1 individually prefers float. Managed float dominates for every sector simultaneously because a positive ϕ_s damps the exchange rate's contribution to inflation variance without fully re-imposing the peg's relocation cost, requiring no sectoral trade-off to still improve on both corner regimes.